

Phishing Attack Detection and Password Analysis System

1. Introduction

With the increasing use of the internet, cyber attacks have become a major concern. One of the most common threats is phishing, where attackers trick users into revealing sensitive information such as passwords, bank details, and personal data through fake emails, messages, or websites.

This project aims to develop a system that can detect phishing attempts and help users identify unsafe content. It also includes a password strength analyzer to encourage users to create secure passwords. The system is designed to improve cybersecurity awareness and protect users from online threats.

2. Abstract

This project presents a web-based application that detects phishing attacks and evaluates password strength. The system analyzes user input such as messages and URLs to identify phishing indicators like suspicious keywords, fake links, and urgency patterns.

In addition, the password analysis feature checks the strength of passwords based on length, complexity, and use of special characters. The application provides feedback and suggestions to improve password security. Overall, the system acts as a simple and effective cybersecurity tool for everyday users.

Tools Used

- **React** – For designing the user interface
- **JavaScript** – For frontend interactivity
- **Node.js** – For backend development
- **NPM** – For managing packages

- **VS Code** – For development and coding

4. Steps Involved in Building the Project

1. **Problem Identification** – Identified phishing attacks as a major cybersecurity issue.
2. **Requirement Analysis** – Defined features like phishing detection and password strength checking.
3. **System Design** – Planned frontend and backend architecture.
4. **Frontend Development** – Built user interface using HTML, CSS, and JavaScript.
5. **Backend Development** – Developed server using Node.js and Express.
6. **Phishing Detection Logic** – Implemented rules to detect suspicious keywords, fake URLs, and sensitive requests.
7. **Password Analyzer** – Added functionality to check password strength and provide suggestions.
8. **Testing** – Tested the system with different inputs to ensure accuracy.

Conclusion

The Phishing Attack Detection and Password Analysis System helps users identify phishing attempts and improve password security. It is a simple yet effective tool that raises awareness about cybersecurity risks. This project demonstrates how web technologies can be used to build practical solutions for real-world problems and enhance user safety online.